

Nishant Gupta

Development Engineer, PhD in Theoretical Physics

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📍 Bengaluru, India

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Summary

AI engineer with a PhD in theoretical physics, building retail AI systems and evaluation workflows, with an active research background in asymptotic symmetries, holography, and celestial/Carrollian conformal field theories.

Education

PhD The Institute of Mathematical Sciences (IMSc), Theoretical Physics Chennai, India
 • Thesis Advisor: Prof. N. V. Suryanarayana 2018 – 2024
 • Research on asymptotic symmetries, holography, and celestial/Carrollian conformal field theories

Experience

eTeam India (Client: Target India), Development Engineer Bengaluru, India
 Building retail automation systems and comparing production architectures on accuracy, cost, and operational fit. Apr 2026 – present
5 months

- Built skills-based retail automation workflows using Google ADK-style architecture for merchandising and planogram generation
- Developed evaluation, testing, and observability workflows to verify constraints and make architecture tradeoffs easier to measure

National Institute of Science Education and Research (NISER), Research Associate Bhubaneswar, India
 Worked on theoretical physics problems through numerical and symbolic computation, resulting in publications, talks, and collaborative research. Mar 2024 – Mar 2026
2 years 1 month

- Built Python and Mathematica workflows for non-linear differential equations and related theoretical physics calculations
- Mentored students and presented research at conferences and seminar series

FlowAI, Physics & AI Expert Remote (India)
 Supported AI work on symbolic mathematics and theoretical physics reasoning problems. Sept 2025 – Jan 2026
5 months

- Helped shape agent architecture choices for hard scientific problem-solving workflows
- Curated benchmark questions for evaluating LLM reasoning quality and hallucination behavior on mathematical tasks

Publications

Worksheet CFT2 and Celestial CFT2: An AdS3-CFT2 Perspective 2025

Preprint.

Banerjee S., Gupta N., Misra S.

(arXiv:2506.14891 [hep-th])

Chiral Λ -bms4 Symmetry of 3D Conformal Gravity 2025

Published in 2025.

Gupta N., Suryanarayana N.V.

(Eur. Phys. J. C 85 (2025) 4, 468)

A Chiral $\mathfrak{A}$-bms₄ Symmetry of AdS₄ Gravity Published in 2025. Gupta N., Suryanarayana N.V. (Nucl. Phys. B 1010 (2025) 116759)	2025
All Chiral W-algebra Extensions of $\mathfrak{so}(2,3)$ Published in 2024. Gupta N., Suryanarayana N.V. (J. High Energ. Phys. 2024, 137)	2024
$\mathfrak{sl}$-hat 2 Symmetry of $R^{1,3}$ Gravity Published in 2023. Gupta N., Paul P., Suryanarayana N.V. (Phys. Rev. D 108 (2023) 8, 086029)	2023
Constructing Carrollian CFTs Published in 2021. Gupta N., Suryanarayana N.V. (J. High Energ. Phys. 2021, 194)	2021

Projects

Autonomous Multi-Agent Data Analysis System Built a multi-agent system for autonomous data analysis, combining statistical EDA, ML modeling, LLM-based explanation, and automated quality evaluation in one workflow. • Stack: LangGraph, Python, LLM evaluation	2025 – 2025
Production RAG System for Python Documentation Built a production-grade RAG application with hybrid retrieval, query rewriting, grounded answer generation, and evaluation for both retrieval relevance and answer quality. • Stack: FastAPI, Pinecone, BM25	2025 – 2025
langchain-community-inspire-hep (PyPI) Built and maintain a Python package that connects INSPIRE-HEP with LangChain so AI applications can search and retrieve high-energy physics literature through typed interfaces and tested tooling. • Stack: LangChain, Pydantic, Python packaging • PyPI: https://pypi.org/project/langchain-community-inspire-hep/	2025 – present

Talks and Presentations

Indian Strings Meeting 2025, IIT and NISER Bhubaneswar , Talk Towards Engineering Celestial CFTs - An AdS-CFT perspective	Bhubaneswar, India Dec 2025 – Dec 2025
CCPP, New York University , Seminar Worksheet CFT2 and Celestial CFT2: An AdS ₃ /CFT ₂ perspective	New York, NY, USA Aug 2025 – Aug 2025
Special Leinweber Seminar, KCTP, University of Chicago , Seminar Worksheet CFT2 and Celestial CFT2: An AdS ₃ /CFT ₂ perspective	Chicago, IL, USA Aug 2025 – Aug 2025
Indian Strings Meeting, IIT Bombay , Talk Chiral W-algebra from 3d conformal gravity	Mumbai, India Dec 2023 – Dec 2023
Indian Institute of Science, Bangalore , Talk All chiral W-algebra extensions of $\mathfrak{so}(2,3)$	Bengaluru, India Oct 2023 – Oct 2023

Skills

AI/ML Engineering

Theoretical Physics

Computational

Languages

English

Fluent

Hindi

Fluent